Bilingual Recipe Translation Using Transformer-based Seq2Seq Modeling

Github*: https://github.com/jackielele/w266-NLP-final*
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Abstract

Food is an integral part of every individual's culture. With technology being readily available at our fingertips, obtaining recipes from various regions in our preferred language has become easier than ever, however many caveats persist when identifying true cultural adaptation of recipes across regions. In this research we aim to create machine translation architectures using the mT5, mT0, and mBART-50 models to translate recipes from English to Chinese, and vice versa, with emphasis on cross-cultural adaptation.

Introduction & Background

Languages are not only an essential part of communication, but they also provide a great avenue to pass down culture across generations. They help us tell stories and provide a way to express our emotions and thoughts. One such intersection of language and heritage are recipes. Recipes vary across cultures, ranging in different languages, an array of ingredients, and numerous techniques; each reflecting its own heritage. These exact cultural nuances however, are not carried over in recipe translations performed by state of the art models today (Mansi et. al, 2024). With this research paper we aim to dive deeper into the cross-cultural construct of recipes and obtain a process to effectively translate between Chinese and English recipes bidirectionally. Furthermore we will leverage the research performed in *Cultural Adaptation of Recipes* (Yong et. al, 2024), and enhance the modeling experiments by building up on existing architectures to create machine translations that are able to capture regional significance. This paper will explore transformer based NLP models to train multilingual datasets that can continue to carry the essence of regional flavors and cooking-traditions.

Data

All models were built on the CulturalRecipe dataset, which contained approximately 45,000 Chinese (simplified) recipes matched to English ones, as well as 45,000 English recipes matched to Chinese. CulturalRecipe consists of raw data from monolingual corpora RecipeNLG and XiaChuFang, where recipe pairs were matched based on title similarity and ingredient overlap (Yong et. al, 2024). Additionally, as translation is not always one to one, on average, each source recipe is matched to 3 different translated target recipes. To prepare the data for Seq2Seq training, all recipes were flattened to 1:1 matches, which resulted in a total of approximately 265,000 recipe pairs. Tokenization was performed following dynamic padding for efficiency and memory conservation. We followed an 80/10/10 split for training/validation/test, as well as additional "human" and "silver" evaluation sets. The human evaluation dataset provided a ground truth to the recipes with translations provided by human experts who translated the recipes with cultural adaptation. The silver test dataset on the other hand was used to provide more automated translation pairs that potentially consisted of noise (Yong et. al, 2024). The raw data was parsed and divided into the following sections: Title, Ingredients, and Steps to be used for model training. Due to the sheer size of the dataset, a smaller subset of 2000 recipe pairs (for each translation direction) were used for fine-tuning and experimentation.

Methodology

Suitable candidates for translation models include encoder-decoder Transformer architectures, specifically pretrained for multilingual generation. While powerful language models such as GPT4 and BLOOM are capable of cross-lingual text generation, in this study, we focused on Seq2Seq Transformer models. The scope was narrowed specifically to mT0, mT5 and mBART-50 since they have been pretrained on substantial English and Chinese corpora, which allows them to preserve semantic meaning during translation. Outside of pure metrics, a goal of our project was genuine cultural recipe adaptation and not only literal language translation. Chinese and English recipes differ not only with grammar and syntax, but also in culturally specific cooking practices: ingredient conventions, measurement systems, preparation styles, etc. Our models strive to adapt ingredients, measurements, and cooking methods that fit the target Chinese or English culture.

mT5-small

For our initial experiments, we selected mT5-small (300M parameters) rather than the mT5-base (580M parameters) model used in the reference paper. Since the mT5-small model requires lower GPU memory and faster iteration while preserving the same architecture, it was a comparable alternative due to our memory constraints. Although mT5-small is trained on fewer parameters, both retain the same overall architecture and vocabulary design, and we did not expect to see a noticeable difference in translation quality. During our experimentation, we first conducted parameter exploration and tuning on a random subset of 2,000 training pairs to get our optimal set of tuned hyperparameters. We then scaled up to the entire dataset (~265,000), which required multiple sessions and saved model checkpoints due to the sheer size of the model and training data. Model selection, experimentation and hyperparameter tunings were performed on Google Colab, but the actual model training and evaluation had to be performed on Kaggle due to GPU memory constraints.

Since training for a single epoch took approximately 12 hours, we opted to train for 3 epochs due to an already shown improved loss. To compensate, the learning rate was increased to 5e-4, which was five times greater than the reference rate of 1e-4. To maximize memory, the Adafactor optimizer was chosen, along with a batch size of 32 (per-device batch size of 2 with a gradient accumulation over 16 steps). However, Fp16, mixed precision training, and label smoothing could not be optimized to conserve memory due to NaN validation loss.

mT0-small

Another part of our methodology was to train the mT0-small model by the Big Science research workshop. Although this model is known for its zero-shot capabilities, we further trained this model with our dataset to examine its performance specific to our research. mT0 is an enhancement to the mT5 model with nearly the same amount of parameters; however, mT0 is further fine-tuned on the xp3 multilingual task mixture (Sumayya, 2026). With mT0's cross-lingual training, and zero-shot capabilities, we anticipated this model to further help improve the translations from our existing improved fine-tuned mT5 model. To identify the configurations of the training parameters we obtained a subset of 2000 flattened bidirectional recipe pairs, each with their source and targets, to train the model with. Once the hyperparameters to tune had been identified (Table 1), we proceeded to train the model. Due to memory resource limitations, we lowered the training size for each of the bidirectional datasets while keeping the 80/10/10 split: 96149 pairs for the training, and about 12000 pairs for the validation and test sets. This enabled a timely completion of training and evaluation for this model.

The key difference in the training parameters between mT0 and our fine-tuned mT5 was the amount of epochs we trained this model for. This led us to effectively compare both the models and as a result, allowed us to set up a fair benchmarking structure. Overall, the aim of this research was to adapt recipes across languages as opposed to merely translating them, therefore we planned on leveraging this model to achieve greater precision and accuracy in the translations being performed.

mBART-50

Unlike the mT models, Meta's mBART-50 model is explicitly designed for cross-lingual generation based on the provided language tokens. Additionally, mBART would be a more ideal candidate for recipe generation as it is pretrained on full document reconstruction, as opposed to span prediction. However, the mBART-50 model was considerably larger (almost 5x) than both mT0 and mT5; consequentially, the model was not well trained due to GPU constraints. Therefore, the per-device batch size was reduced to 1 and gradient accumulation increased to 32 steps. Positively, there was a noticeable loss drop with each loaded checkpoint. The model also required careful tokenization setup: the src_lang attribute was explicitly set to either zh_CN for Chinese, or en_XX for English, prior to encoding the source since explicit source and target language tokens were required. Additionally, we created separate tokenization functions for each bidirection. Although the mBART-50 model was trained for 3 epochs, the model could only be fully trained once and not be further optimized due to errors, session time-outs, and GPU limitations, as a 12 hour session could only train half an epoch, requiring checkpoints. The default mT5 hyper-parameters were also kept for mBART-50 since the sharp drop in loss was promising; such as, the high learning rate of 5e-4 to compensate for the low number of epochs.

	Baseline (paper -mT5 base)	mT5 (small)	mT0 (small)	mBART-50
Batch Size	32	32	32	32
Epochs	30	3	2	3
Learning Rate	1e-4	5e-4	5e-5	5e-4
Optimizer	Adam	Adafactor	Adafactor	Adafactor

Table 1. Hyperparameter Configuration for each model

Note: While 2 40GB A100 GPUs were used to train the paper's baseline mT5 model, only cloud T4 GPUs were used to train the mT0, mT5-small, and mBART-50 models.

Results

Our main evaluation metrics are BLEU and ROUGE, which is typical for multi-lingual translation models. Our improved mT0 and mT5 models both resulted in higher evaluation scores overall compared to the baseline model described in *Cultural Adaptation of Recipes.* The mBART-50 model performed better or comparable in the cn2en direction, but drastically worse in the en2cn direction. We had two different evaluation datasets we ran the BLEU and ROUGE metrics on (silver vs. human), and this included evaluation for both directions of language translation. These results are further outlined in Table 2, which displays the metric scores across all the models including the baseline paper model. Interestingly, although both mT0 and mT5 models produced the highest scores for the silver_en2cn dataset, the paper's mT5-baseline model outperformed all, suggesting our models struggle with English to Chinese generations and require more training.

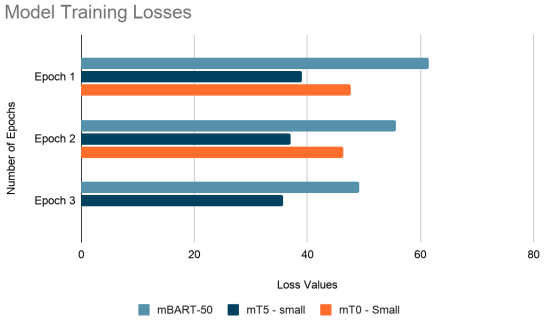
		mT5-baseline	mT5-small	mT0-small	mBART-50
human_cn2en	**BLEU**	1.60	4.19	4.94	4.30
	ROUGE1	25.26	30.39	28.68	28.23
	ROUGE2	4.29	6.41	4.97	5.41
	ROUGEL	17.84	22.39	20.25	20.70
human_en2cn	**BLEU**	15.2	11.68	1.80	7.07
	ROUGE1	32.97	35.66	28.55	20.35
	ROUGE2	11.93	13.43	12.37	3.11
	ROUGEL	31.69	33.49	27.73	16.68
silver_cn2en	**BLEU**	3.74	3.94	4.5	3.86
	ROUGE1	28.93	29.31	28.89	27.42
	ROUGE2	6.92	7.69	7.59	5.95
	ROUGEL	19.51	20.91	21.30	19.58
silver_en2cn	**BLEU**	16.71	11.19	9.01	6.72
	ROUGE1	48.31	47.92	45.87	12.23
	ROUGE2	22.71	18.15	22.09	0.53
	ROUGEL	45.72	44.69	45.16	11.37

Table 2. BLEU & ROUGE Metrics for bidirectional translations

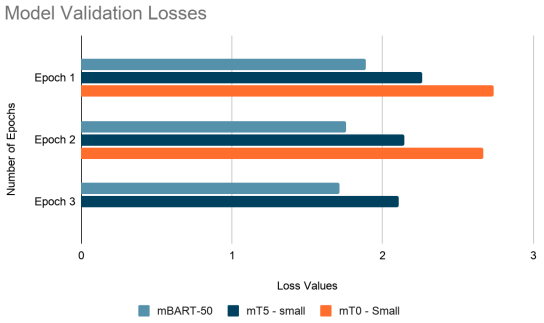
To evaluate the ROUGE and BLEU metrics, we standardized a decoding function across all models using a unified beam-search algorithm (num_beams = 5). This produced the most stable recipe generations without target token-forcing and provided a strong evaluation algorithm that measures cultural adaptation ability, not just explicit language-token translation. Other parameters such as no_repeat_ngram_size and repetition_penalty were also trystormed with in order to find the optimal value that would be best for both English to Chinese translations and vice versa. In addition to the BLEU and ROUGE scores, each model showed improved validation and training loss for each sequential epoch. The final validation loss for the mT5 model on 3 epochs was 2.1 and mBART was 1.7, while the mT0 model had a validation loss of 2.6 at the end of 2 epochs. Graph 1 further demonstrates the training and validation loss comparisons for the mT0, mT5, and mBART-50 models.

Model: mT5-small (Google)			**Model: mT0 (Google)**			**Model: mBART-50 (Meta)**		
Epoch	**Training Loss**	**Validation Loss**	**Epoch**	**Training Loss**	**Validation Loss**	**Epoch**	**Training Loss**	**Validation Loss**
1	39.073	2.260	1	47.609	2.736	1	61.556	1.891
2	36.971	2.142	2	46.355	2.668	2	55.599	1.760
3	35.686	2.103	N/A	N/A	N/A	3	49.215	1.716

Table 3. Training & Validation Loss Improvement for 3 finetuned models



Graph 1a. Training loss for each model



Graph 1b. Validation loss for each model

Discussion

In Table 2, the highest evaluation score is highlighted in green for each of the different evaluation metrics (BLEU or ROUGE) for each translation direction. The English → Chinese translations were more difficult for our fine-tuned models, likely due to the more complex Chinese syntax and lexical compressions. In addition, the silver set references are more "formulaic" (similar to training-style outputs) which would inherently raise the metrics. The human matched sets are more diverse, leading to harder lexical matching.

However, for Chinese to English translations, the fine-tuned mT5-small consistently performed the highest for all BLEU and ROUGE metrics. Compared to the baseline, this is most likely attributed to the higher learning rate (5e-4 vs. 1e-4) which allows for faster initial adaptation to the recipe domain, a trade-off for fewer epochs. mT5-small likely generalizes well to structured recipe generations due to its pretraining objective. Chinese to English may also be inherently easier for the model, as there is strong English pre-trained and generation capabilities. Additionally, research has shown that for Seq2Seq tasks, the Adafactor optimizer outperforms Adam in domains such as text generation or large-scale modeling optimization, in addition to memory overhead reduction. (Zhao et al, 2024). However, mT5-small underperformed in the en2cn for the silver-set and certain metrics in the human set, although some results were still comparable. The BLEU gap for the silver dataset is especially notable. Factors of this lack of precision may be due to BLEU's requirement for close to exact n-gram matches, whereas the Chinese language has more different exchangeable characters. However, the ROUGE scores, although slightly lower, were quite comparable, insinuating that even for the English to Chinese translations, the generation may be relevant but diverge from the reference.

The training and validation loss of the mT5-small model, shown in Table 3, shows an early gain from epoch 1 to 2, where the validation loss drops by 0.118, indicating that our model has learned the basic structure of recipe adaptation. However, there is a plateau in the drop from epoch 2 to 3 (0.039), which could mean that the model is hitting a ceiling from the training data provided by this high learning rate. Our 3-epoch trained model most likely did not fully converge, but was also not overfitted on more data and noise.

The fine-tuned mT0-small model performed well overall, wherein it produced outputs whose BLEU and ROUGE scores were improved from the baseline model for certain translation directions. Of the two evaluation sets, the mT0 model outputs rendered better metrics on the silver test datasets. Improved scores on the silver datasets can be attributed to the use of the automated target values produced by models as opposed to the human evaluated targets. This can lead to higher scores for the silver dataset since the output and target would both be machine translations without any human influence. The model also seemed to demonstrate higher ROUGE scores compared to the BLEU scores. This shows how the model translations were able to carry the context of the overall target recipe translations. Hence, although the BLEU metric underperformed, demonstrating a lower precision for exact translation matches, the model rendered recipes that closely aligned in semantics with the target adapted translations. Intriguingly, the mT0 model outperformed the models in some of the BLEU scores, indicating that this model may have been able to provide better character matches for the Chinese to English translations compared to the other models.

The training and validation loss for mT0 showcased greater stability with the loss consistently decreasing across the two epochs. With checkpoints being saved frequently for this model, evaluation for each checkpoint was also conducted, and it indicated that the training and validation loss had started to converge by the end of the second epoch while both losses continued to decrease. Given the limited

amount of processing capacity, two epochs would not aid in accurately deciphering this model's generalization capabilities, and therefore additional epochs would need to be trained to obtain a more accurate representation of this model's generalization. We anticipate with more epochs and through training on the entire dataset, this model may have the ability to improve in both the loss and the metrics results as well.

Although the mBART model infrastructure is more catered to our translation generation objective, our mBART model drastically underperformed, especially for the cn2en datasets, where it was the worst for all metrics. This was most likely due to decoding sensitivity and incomplete direction convergence for English → Chinese translations; mBART failed to successfully generate target Chinese recipe translations, only English ones. Our model is highly sensitive to decoding policy and insinuates a decoder-side Chinese generation issue. Training for more epochs with a lower learning rate could result in a more stable convergence and improved Chinese generation. Additionally, mBART is generally stronger at mirroring literal translations rather than cultural adaptation, especially in comparison to the mT family that leverages contextual understanding. However, we are confident that given more time and resources to optimize, our model will head towards convergence, as it was discovered during model evaluation that decoding without using explicit language steering yielded improved, as well as representative results.

Conclusion

Overall, the mT5-small model outperformed the baseline mT5-base, fine-tuned mT0-small, and mBART-50 on Chinese to English translations, with the mT5-baseline performing better on the English to Chinese translations amongst the four models. This is promising since it shows that smaller and less trained models can still generate meaningful translations without requiring an abundance of resources, while still maintaining adequate cultural adaptation due to the mT5 pre-trained objective.

We decided to use the BLEU and ROUGE metrics to evaluate the models to effectively compare improved results to what was reported in *Cultural Adaptation of Recipes,* however, we believe that a more accurate evaluation strategy could be set up that checks for cultural constraints through proper tokenization for food substitutes which would add to the BLEU and ROUGE metric scores. This would be able to provide for a more accurate metric that can be leveraged to further test for cultural adaptiveness along with recipe translations.

Since the mBART-50 is more specifically designed for cross-lingual generation, further training and optimization of this model, given more GPU memory, would be beneficial for meaningful improvement. Improved decoding on the Chinese generation direction would also be required. Specifically, separate models could be trained for each direction with varied learning rates. Additionally, given limited existing international language datasets, with more processing capacity and for a longer term project, creating a parallel dataset using machine translation and running these models on that would also enable us to experiment with other languages.

Contributions

Jackie contributed to the following sections of the paper with a focus on the mT5 and mBART-50 model training components: Data, Methodology, Results, Discussion, Conclusion

Vrushti contributed to the following sections of the paper with a focus on the the mT0 model training components: Abstract, Introduction, Methodology, Results, Discussion, Conclusion

References

1. Cao, Yong, et al. *"Cultural Adaptation of Recipes." Transactions of the Association for Computational Linguistics*, vol. 12, 2024, pp. 80–99. MIT Press, https://doi.org/10.1162/tacl_a_00634.
2. Goel, Mansi, et al. *"Deep Learning Based Named Entity Recognition Models for Recipes." Proceedings of the Language Resources and Evaluation Conference (LREC-COLING 2024)*, 20–25 May 2024, pp. 4542–4554. European Language Resources Association (ELRA), https://aclanthology.org/2024.lrec-main.406.pdf.
3. Soltan, Saleh, and Andy Rosenbaum, et al. *"Recipes for Sequential Pre-Training of Multilingual Encoder and Seq2Seq Models." Findings of the Association for Computational Linguistics: ACL 2023*, 2023, pp. 9380–9394, https://doi.org/10.18653/v1/2023.findings-acl.598.
4. Negru, Vlad-Andrei. *"From Text to Taste: Advancing Smart Appliances with Multilingual Recipe Interpretation." 2024 IEEE 40th International Conference on Data Engineering Workshops (ICDEW)*, https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10555101.
5. Afreen, Sumayya. "Evaluating Multilingual Models: A Comparative Evaluation of MT0 and MT5 for Urdu Text Articles Summarisation." *Journal of Electrical Systems*, journal.esrgroups.org/jes/article/view/5804. Accessed 11 Apr. 2026.
6. Naveen, P., and Pavel Trojovský. "Overview and Challenges of Machine Translation for Contextually Appropriate Translations." *iScience*, vol. 27, no. 10, 2024, p. 110878. Elsevier, doi:10.1016/j.isci.2024.110878.
7. Tanwar, Neelesh Kumar, Atul Joshi, and Ankur Singh Bist. "A Comparative Study of MBart and Alternative Transformer Models for Kumauni Language Translation." *Asian Journal of Mathematics and Computer Research*, vol. 32, no. 3, 2025, pp. 147–161. doi:10.56557/ajomcor/2025/v32i39482.
8. Cruchinho, Paulo, et al. "Translation, Cross-Cultural Adaptation, and Validation of Measurement Instruments: A Practical Guideline for Novice Researchers." *Journal of Multidisciplinary Healthcare*, vol. 17, 2024, pp. 2701–2728. *PubMed Central*, doi:10.2147/JMDH.S419714.
9. Zhu, JiaSheng et al. "Improving multilingual English translation performance through T5 and MAML integration" https://doi.org/10.1016/j.sasc.2025.200394
10. Zhao, Rose, et al. "Deconstructing What Makes a Good Optimizer for Language Models" *Harvard University, arXiv:2407.07972v1*